

Week 02 Task 6

Understanding Logs & Log Analysis Basics

CandidateName: Asad Ali
Reference Number: TBS-INT-2026-445
Submission Date: 27|10|2026
Submitted To: TechBiz Security

Table of Contents

1) Executive Summary 3
2) Types of Logs 3
    2.1) How logs are being Monitored 4
3) Apache web Server log line 4
4) Difference between Suspicious and Normal Log Entry 5
    4.1) Example of Malicious Log Vs Normal Log 5
5) Baseline in Log Analysis 5
    5.1) Purpose of Baselining 6
    5.2) Key Metrics for Establishing a Baseline 6
6) TryHackMe Room Intro to Logs 7
7) TryHackMe Investigating Windows 7
8) Conclusion 8

1) Executive Summary

This task focused on developing a practical understanding of logs and basic log analysis in cybersecurity. It covered different types of logs, their sources, common log locations, and the role of centralized log monitoring through SIEM platforms. A practical Apache web server exercise was also performed by generating an HTTP request with **curl** and analyzing the resulting access log to identify key information such as the source IP address, timestamp, HTTP request, response status code, and response size. The task also focused on distinguishing **normal** and **suspicious log** activity through behavioral patterns and security baselining. I learned how establishing a baseline of **normal authentication, network traffic, process, file access,** and log-generation activity can help identify anomalies and reduce false positives. In addition, practical exercises on TryHackMe's Intro to Logs and Investigating Windows rooms provided hands-on experience with log analysis, Windows artifacts, user activity, suspicious files, listening ports, persistence mechanisms, and indicators of compromise.

2) Types of Logs

Logs are chronological records of activities and events generated By operating systems, applications, server and Network devices. They provide comprehensive visibility about System Operations such as user activities, authentication attempts, network communications and potential security incidents. A log may contain timestamp, event description source , user account, IP address event identifier and other relevant details related to event.Logs are important for troubleshooting,Security Monitoring, threat detection, incident investigation There are Many Types of Logs some of them are mentioned below.

1. **System Logs:** Generated by the operating system to track core events, such as system boot ups, shutdown events, hardware changes, driver errors, and service states.

Origin of System Logs: Operating system compoents , kernel, device drivers, and hardware related events including involving componets like RAM and CPU.

File Path: For windows File Path is **C:\Windows\System32\winevt\Logs\System.evtx** and For linux File path is **/var/log/sys.**

2. **Security Logs:** Focus on authentication and access control, recording successful/failed logins, privilege escalations, and account modifications.

Origin of System Logs: They come from the OS Local Security authority (LSA), Active Directory Domain Controllers, authentication daemons (sshd), and system audit frameworks (auditd).

File Path: For windows File Path is **C:\Windows\System32\winevt\Logs\Security.evtx** and For linux File path is **var/log/auth.log (or /var/log/secure)**

3. **Application Logs:** Generated by software applications to track runtime errors, database queries, user actions, and application-specific events.

Origin of Application Logs: They come from applications and software, databases (MySQL, PostgreSQL), custom enterprise applications, and programming runtime environments (Node.js, Python, Java JVM). on the Operating System

File Path: For windows the File Path **C:\Windows\System32\winevent\Logs\Application.evtx** and for linux the File Path **/var/log/app_name.**

4. **Network and Firewall Logs:** Network Logs Capture traffic patterns, routing data, and connection attempts across network devices like routers and switches while Firewall logs commonly tracks connections and traffic that are allowed or blocked according to firewall rules.

Origin of Application Logs: They come from physical and virtual network infrastructure devices, including routers, switches, wireless access points (WAPs), and VPN gateways.

File Path: For windows the File Path is **C:\Windows\System32\LogFiles\Firewall\pfirewall.log** and for Linux the File path is **/var/log/ufw.log**

5. **Web Server Logs:** Record incoming HTTP/HTTPS requests, client IP addresses, requested URLs, user agents, and response status codes (e.g., 200, 404, 500).

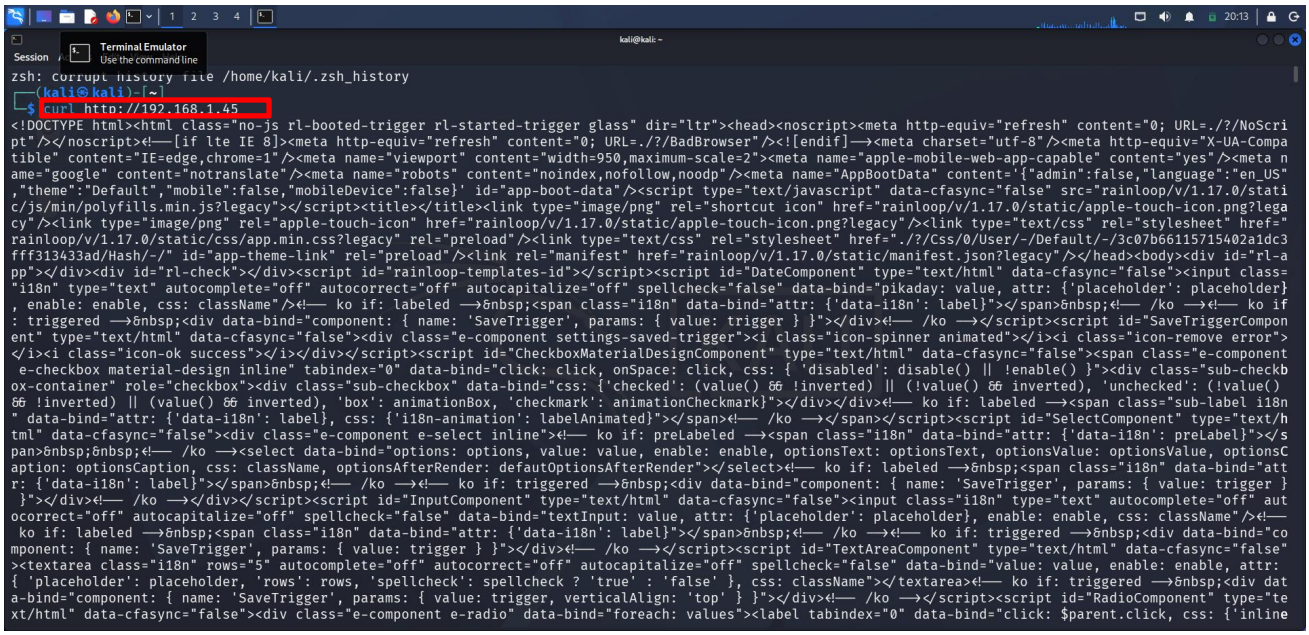
Origin of Application Logs: They come from web server daemons and reverse proxies such as Apache HTTP Server, Nginx, Microsoft IIS, and HAProxy.

File Path: For windows the File Path is **C:\inetpub\logs\LogFiles\W3SVC1** and for Linux the File Path is **/var/log/apache2/access.log**

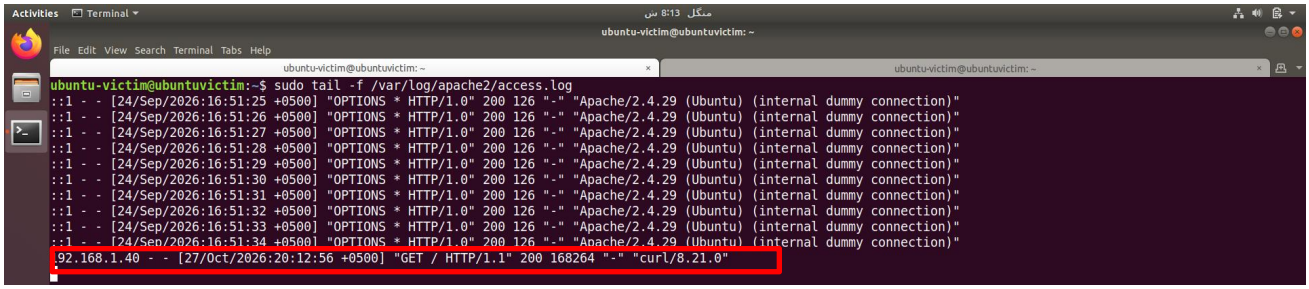
2.1) How logs are being Monitored

In a production environment, individual logs are rarely checked manually on each machine. Instead, they are shipped to a centralized SIEM **Security Information and Event Management** platform like **Splunk, IBM QRadar , Wazuh** where they are parsed, correlated, and monitored for automated threat detection and alerting.

3) Apache web Server log line



In the above figure, a curl command was used to send an HTTP request to the Apache web server. The purpose of generating this request was to create an access log entry and observe the Apache log format.



In the above figure it is shown that as we made a curl request to web server then a log gets generated immediately.

Log Breakdown

Field	Example	Meaning
IP Address	192.168.1.40	Client that generated the request
Time	27/oct/2026:20:12:56	When Apache received it
Request	GET /HTTP/1.1	HTTP method, requested resource, and protocol
Status	200	Request completed successfully
Size	168264	Response size in bytes

4) Difference between Suspicious and Normal Log Entry

Normal Log	Suspicious Log
Logins or actions happen during standard business hours	Login attempts or process executions happen at unusual or unexpected hours
Single or low-frequency requests match a user's typical daily routine or predictable automated schedules (like scheduled backups	Shows repeated failures followed by a sudden success possibly brute force , impossible travel logins from two distant cities in minutes, or new processes appearing without change tickets
Actions succeed cleanly or throw standard, expected system messages	It breaks the behavioral pattern of the specific user or system rather than just matching a static rule.

4.1) Example of Malicious Log Vs Normal Log

Normal Log: Access to device comes from familiar IP addresses, Normal routine timing, known countries, and regular devices.

Malicious Log: Access to device originates from unexpected foreign countries, rare IP blocks, or unknown VPN/Tor nodes.

5) Baseline in Log Analysis

Baselining is a cybersecurity technique where administrators and SOC analysts identify, track, and document the normal, day-to-day behavioral patterns of an organization's digital environment. By continuously monitoring routine operations such as **standard device IP addresses**, typical **login times**, and regular volumes of **incoming and outgoing network traffic** analysts establish a definitive benchmark of business as usual.

5.1) Purpose of Baselining

The main Purposes of Baselining are mentioned below

1) Anomaly & Threat Detection: Helps analysts immediately identify unusual activities such as unexpected login attempts at midnight, suspicious data transfers by comparing live traffic against established normal behavior

2) Reduce false Positive: Reduces security alert fatigue by filtering out expected, routine network behavior so analysts can focus on actual security incidents.

3) Support incident investigation: Provides clear visibility into what normal system behavior looks like, allowing SOC teams to quickly isolate affected systems and contain threats during an active attack.

5.2) Key Metrics for Establishing a Baseline

To create an accurate security baseline, SOC analysts typically monitor and analyze the following key metrics across logs:

1) **Authentication & Login Patterns:** Standard login hours, failed vs. successful login ratios, unusual geographic locations (IP addresses), and user privilege escalations.

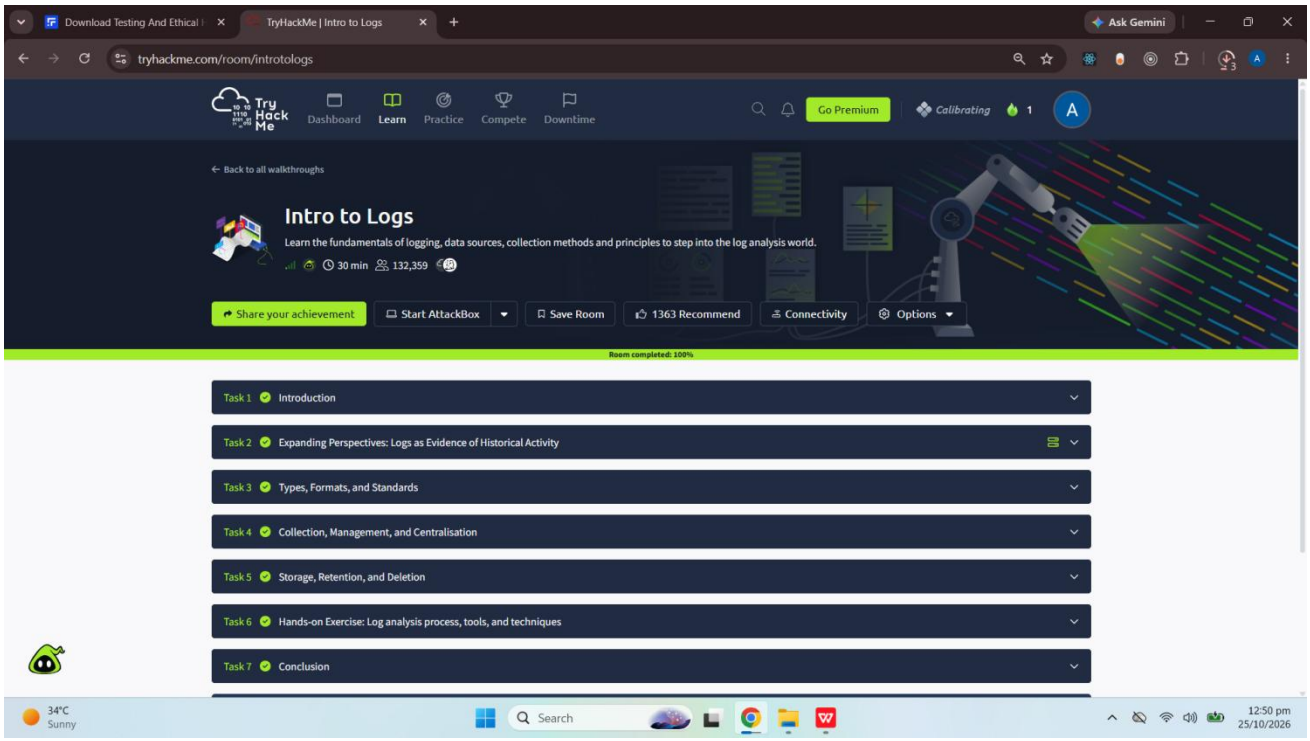
2) **Network Traffic & Bandwidth:** Typical incoming/outgoing data volumes, protocol usage (e.g., HTTP, DNS, SSH), standard port usage, and internal lateral movement.

3) **Process & System Activity:** Normal CPU/memory usage, standard process execution lists, script execution patterns (e.g., PowerShell, Bash), and registry modifications.

4) **File Access & Data Transfer:** Regular file access frequencies, sensitive directory interactions, and bulk file download/exfiltration indicators.

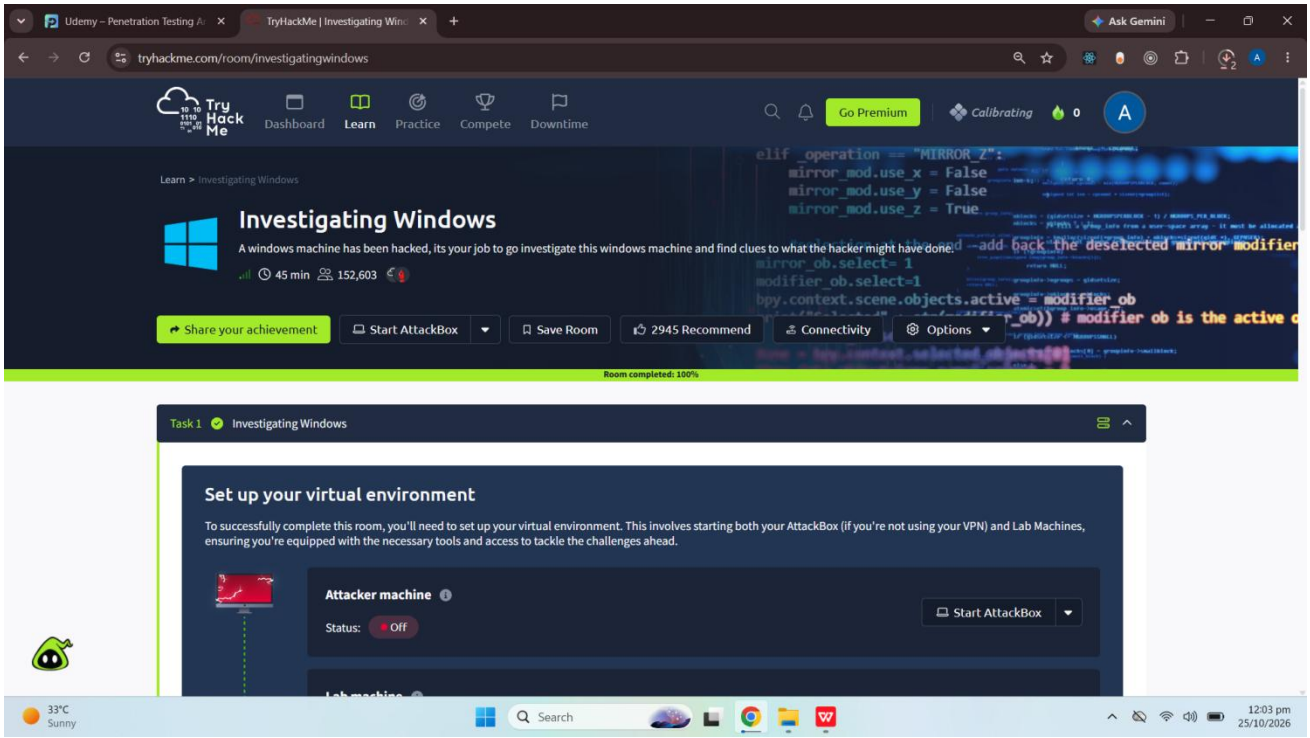
5) **Log Volume & Generation Rates:** Daily log volume baselines per source device to detect log suppression or log flooding attacks.

6) TryHackMe Room Intro to Logs



In the above figure it is shown that I completed the **Intro to Logs** room on TryHackMe, which provided a practical introduction to log analysis. The room covered different **log types, formats, and standards**, along with methods used for **log collection, management, centralization, storage, and retention**. I also learned how logs can be used as evidence of historical system and user activity. The hands-on exercises helped me understand basic **log analysis processes, tools, and techniques** used to identify important or suspicious events.

7) TryHackMe Investigating Windows



In the above figure it is shown that Investigating Windows Room on TryHack Me is completed.where I investigated a compromised Windows machine to identify system information, user logon activity, scheduled tasks, administrative accounts, suspicious files, listening ports, and attacker activity. This provided practical experience in analyzing Windows artifacts and logs to trace attacker actions, identify persistence mechanisms, and gather indicators of compromise (IOCs).

8) Conclusion

This task helped me develop a better understanding of log analysis and its importance in cybersecurity monitoring. I learned about different types of logs, including security, system, application, and network logs, and understood where these logs are generated and how they can be useful during security investigations. I also learned how to differentiate between normal and suspicious log entries by understanding baseline activity and identifying unusual behavior. Additionally, analyzing Apache web server logs helped me understand the structure of log entries and how information such as IP addresses, timestamps, HTTP methods, requested resources, and status codes can be used for investigation.